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To Turn Utilities
Smart

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1 POCKET-SIZED DEVICE To Turn Utilities Smart

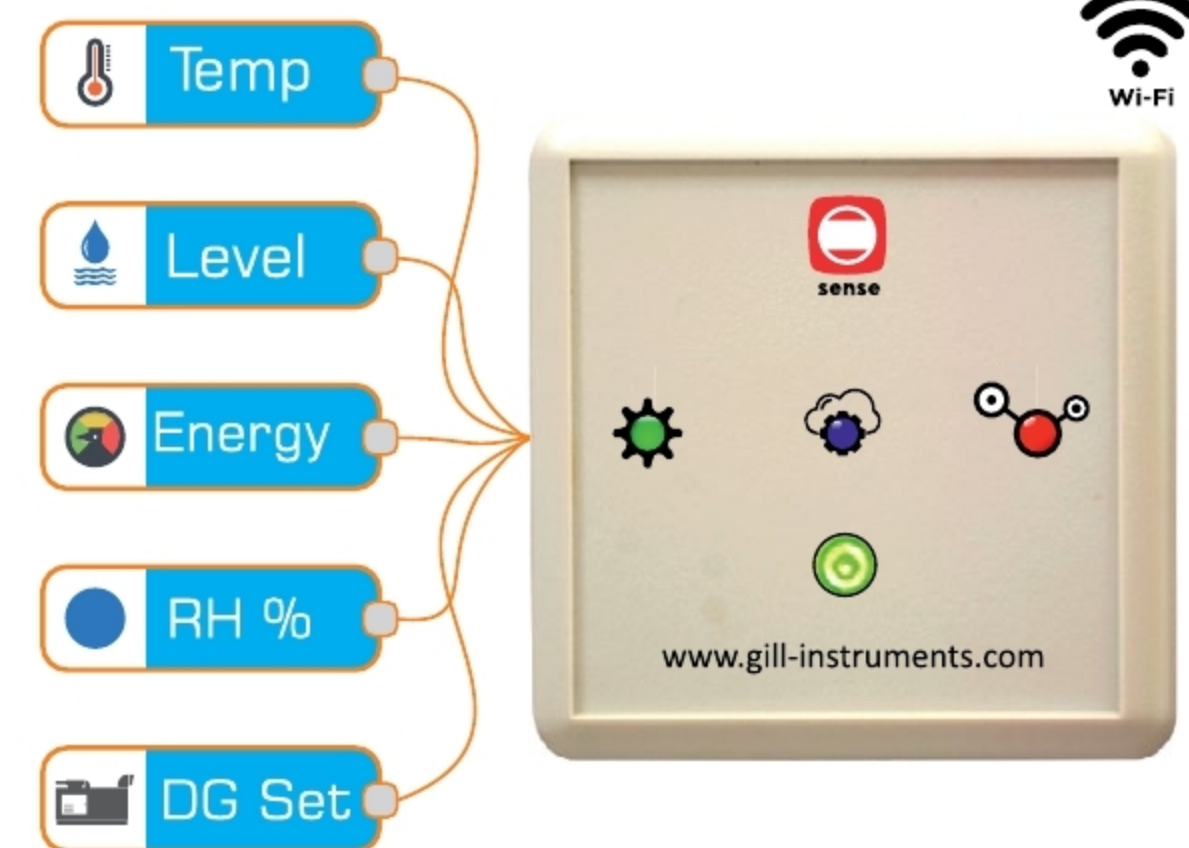
Depleting natural resources, such as fuel and water, have raised a lot of eyebrows. To provide immediate remedial actions, innovators are leveraging on technology to reduce resource wastage. Bengaluru-based Gill Instruments has created a device called Gill Sense, which enables industries, buildings as well as cities to measure readings of energy meters, water meters, diesel generator sets and similar appliances, in real time.

Gill Sense Wi-Fi gateway is a portable device that connects with appliances such as water meter, electric meter, diesel generator set, tank level gauge, temperature, pressure and humidity sensors, and so on, to store their data in the cloud, and produce real-time analytics of these.

Gurjit Singh, founder and managing director, Gill Instruments, says, "The device follows a three-fold approach—sense, connect and manage. This approach fits with any smart premise, be it an industry floor, smart building, smart city or even healthcare unit. All kinds of utility and environmental parameters can be measured with this product.

"In the first step, Gill Sense takes inputs from all such sensory devices. In the second step, it uses Wi-Fi to establish connectivity with the cloud. We have partnered with Amazon Web Services, where data is stored securely and processed. In the final step, data is processed, analysed and managed on the cloud. A software application displays the data on a dashboard."

Gill Sense is designed using widely-known industry standards for operating. For instance, it uses Modbus protocol for energy management. For



water meter readings, it incorporates RS-485 meter standard. Other standards include RS-232, PWM and more.

As the device is standardised, it is compatible with meters from any manufacturer or vendor, who also complies with the set standards. It enables connectivity of legacy meter devices as well.

Using Gill Sense mobile application, users can access their consumption data remotely. Singh explains, "Gill Sense can also help identify any system malfunction. For instance, let us say, a factory's daily energy consumption pattern is around 100 units per day. If suddenly, for a couple of days, it goes up to 200 units, the owner(s) can take immediate action to find the source of excess energy consumption and take corrective action immediately. The app can also generate alerts and send notifications

in such cases."

Gill Instruments has won a partnership with Electronic City Industrial Township Authority (ELCITA), Bengaluru. Gill Sense is being deployed across Electronic City Township for smart city applications. The Gill Sense team has an office there, where it monitors the deployments in real time.

Gill Sense comes with a product-cum-service model. Singh explains, "Cost for the service depends on the scale of requirement. Some customers may require only the mobile/desktop application, while others may want the APIs, too, to access data and customise their own analytical dashboard and interface. Cost will depend on all these considerations."

The hardware is a one-time purchase, with different costs for application-wise variants. For instance, hardware unit for water meter monitoring